

### **Abstract**

The aim of this sub-project is to develop an image processing component to support low-cost automated greenhouses. The image processing application must enable the greenhouse system to establish various aspects about the plants from the images taken. These aspects include plant health, plant growth and plant disease identification from the images of different parts of the plants in the greenhouse.

**What is the purpose of this section?**

- Context and importance of your research.
- Outline of literature review

**Why choose image processing techniques?**

- Former methods cost too much time and were expensive + destructive [12]
- Focus on image analysis component [5]
- Review plant phenotyping concepts in relation to image processing and plant breeding, discussing various measurable traits

# 1 Plant Phenotyping

Plant breeding is defined as the process of inducing desirable inheritable traits in plants through controlled human action [19]. This process is important in solving the problem of adequate global food supply. The rapid growth of the human population has created a strain on global food supply. It is estimated that crop production must increase by 70% by 2050 [3] to meet this demand. Additionally, abiotic and biotic stressors limit cultivar biomass and yield [7]. Therefore, plant breeders need effective techniques to produce germplasm for cultivars that maximise yield and biomass, in various adverse environmental conditions. In the last few decades, image processing techniques in plant phenotyping have been shown to be an effective tool for plant breeding [1].

## 1.1 What Is Plant Phenotyping?

Plant phenotyping is the evaluation of phenotypes in plants [6]. Phenotypes are primitive or complex traits of an organism that can be observed or quantified and evaluated numerically [10]. The literature also indicates that phenotype can alternatively be seen as combination of all morphological, physiological, chemical, and behavioral aspects that represent the individual plant [8, 18, 15].

## 1.2 Plant Phenotyping As Method For Improving Plant Traits

Quantitative data produced by plant phenotyping is most useful in analyzing plant-environment interactions in plant breeding [6] for development of germplasm with improved plant traits [15]. Phenotypical data is useful as a proxy for assessing plant performance in relation to its environment. For example, stressors or disturbances of plants are induced by environmental conditions that organisms are placed in, and these stressors induce symptoms which can be detected and evaluated using phenotyping. This indicates phenotypical traits of a plant have complex dynamic feedback loop with their environment [28], and therefore phenotypical traits serve as proxies in determining if certain genotypes have favourable traits in particular environmental conditions.

## 1.3 Image-Obtainable Phenotypical Traits As Indicators Of Plant Health, Growth, And Stress

Phenotypical traits that can be obtained using image processing techniques can be divided into morphological, physiological, and temporal traits. Additionally the literature indicates, morphological and physiological measurements are classifiable as holistic-based or component-based [4]. Holistic traits consider the plant as a whole, whereas component-based features look at particular organs like the fruit, flower, leaves, stem etc. of the organism.

**Leaf phenotypic traits**

Leaf-derived traits are one of the most commonly used type of traits, because leaves rapidly respond to environmental stimuli, and robustly indicate various factors which determine plant health, growth and stress. [30]. Leaf phenotypic traits have been used to predict plant health, growth and stress. For example, the morphological trait of leaf area predicts plant productivity, because reduced leaf area results in decreased chlorophyll content, and consequently less photosynthesis. Therefore leaf area is a good proxy measure for evaluating plant growth, and has been used to monitor and estimate crop yield, and health [31].

**Morphological traits** About plant structure, which affects its performance and gives indication regarding its health

- Leaf width
- Number of leaves
- Leaf inclination angle
- Leaf area
- Root system architecture (e.g root diameter)
- Plant shoots
- Fruit
- Flowers

**Physiological traits** About how plant operates in response to its growing conditions

- Stomatal conductance
- Chlorophyll content of leaf
- Plant water status
- Plant photosynthesis (metabolism)
- Nutrient content

**Phenological traits** These are phenological traits (when certain biological events occur that depend on climates)

- When fruit or flowers are yielded
- Yield prediction

**The use of plant phenotyping and detecting plant stress** Plant biomass is affected by stresses and disturbances [7]. A stressor is anything which stymies the growth of plant biomass. Examples of stressors include things like water scarcity, inadequate amounts of light, and unfavourable temperatures. A disturbance is anything which causes some level of destruction to the plant. Examples of disturbances include pathogens, fires, frost, and pests

**Biotic Stress + Disturbance.**

- Pathogenic organisms
- Pests
- Parasitic plants

**Abiotic Stress + Disturbance**

- Drought
- Flood
- Salinity
- Heat Stress
- Radiation Stress, light stress

### **The Relationship Between Image Processing And Plant Phenotyping**

Plant phenotyping methods with image processing are non-destructive, amenable to automation, and readily scalable to high throughput scenarios in large samples. This is in stark contrast to traditional phenotyping methods which are prone to human error, extremely time-consuming, costly, and are difficult to perform from start to end of a plant's ontogenesis [2, 30, 4].

#### **What is the purpose of this section?**

- Discuss image processing in the context of plant phenotyping
- Discuss image acquisition techniques
- Discuss image processing techniques used in plant phenotyping

#### **How does image processing relate to plant phenotyping?**

Fill out this section

### **The conceptual pipeline for image processing techniques in plant phenotyping**

This paragraph should discuss how image processing is applied. I.e it should map the process from image acquisition to image processing to data generation/analysis

#### **Foundations of image acquisition in plant phenotyping**

- measure a phenotype quantitatively through the interaction between light and plants such as reflected photons, absorbed photons, or transmitted photons. Each component of plant cells and tissues has wavelength-specific absorbance, reflectance, and transmittance properties [15]
- For example, chlorophyll absorbs photons primarily in the blue and red spectral region of visible light [15]
- Imaging at different wavelengths is used for different aspects of plant phenotyping [15].
- Imaging techniques are very helpful for detecting these properties, especially for those that cannot be seen by the naked eye [15]

#### **Image acquisition methodologies in plant phenotyping**

##### **Thermal imaging sensors**

- Conceptual basis of this technique [21]
- Have been used to measure plant water status.
- Used in Irrigation scheduling.

- By inferring transpiration rate of leaves which is governed by stomatal conductance.
- Stomatal conductance can indicate drought, pathogens, and also soil salinity.
- Have also been used for yield predictions [9]

### **Hyperspectral sensors**

- Based on the principles of spectroscopy. Can detect biochemical, morphological and physiological traits of plants. Has been used to detect drought stress. Also used to detect water stress, and pathogens. Different reflectance can tell us about different things. Certain band tells us about water stress, some can tell us about chlorophyll content. [24].
- Conceptual basis and applications [23]
- Crop yield
- Biotic and abiotic factors

### **Tomography**

- three-dimensional visualization of plant internal structures, root systems, and physiological processes
- Based on principles of x-rays, moving sensor takes planar cross sections to 3D visualize plant
- Drought stress [27]
- Yield Estimations
- has references for usages [20]

### **Fluorescence sensors**

- Conceptual basis of this technique [25]
- Predict metabolism using photosynthesis which tells us about growth status
- Any stress or disturbance can indirectly or directly affect plant metabolism, so chlorophyll content can be used as a measure to detect these things like pathogens, light shortage, nitrogen deficiency, water stress etc [16]
- In optimal conditions, very little heat is emitted in photosynthesis so in certain near infrared emission should be low
- Actinic light involved here?

### **RGB Camera**

- Mimic human vision
- measure plant shoot biomass (yield potential) [15]
- leaf senance (tissue resistance to salt)

### What is the purpose of this section?

- Discuss how image processing and plant phenotyping are applied in low-cost automated greenhouses
- What methods, limitations and advances have there been.
- What does a typical image preprocessing pipeline look like?

### Basis for future in low-cost imaging analysis

The basis for the points below is from [nabwireReviewApplicationArtificial2021]

- Existing phenotyping systems are commercially developed and under patent, so are not modifiable or readily available [13]
- RGB, and hyperspectral sensors most commonly used as they are affordable
- Data collection and analysis tools are not generalizable or standardised, so many custom flows are made
- Indicating the need for standardised, extendable image analysis pipelines using readily available open-source software and hardware
- Recent advances show the use of cheap hardware like raspberry pi, and affordable RGB sensors [26]
- However these sensors don't have a lot of compute power, so analysis and preprocessing is done remotely
- Cheap sensors but are cheaper and faster
- Cloud architectures exist, but with amount of data generated this cost can be prohibitive in terms of processing cost [bauckhageDataMiningPattern2013]. Cyberinfrastructure expensive ref (95,96)
- Moreover, existing systems are not designed for multiple genotypes so their applicability is limited, suggesting need for more generalizable image analysis
- Deep learning and machine learning can be used to automate image analysis, and reduce costs
- Availability of large labelled data is bottleneck
- To use on cheap hardware you also need to optimize
- Therefore low complexity efficient algorithms for automated image preprocessing and image analysis is needed
- Using multimodal segmentation is also another angle

- ref[36], one of the advantages of using ML approaches in plant phenotyping is their ability to search large datasets and discover patterns by simultaneously looking at a combination of features (compared to analyzing each feature separately). This was previously a challenge because of the high dimensionality of plant images and their large quantity, making them difficult to analyze through traditional processing methods (ref 37)
- machine learning achieved using supervised learning, so we are not able to detect novel or unexpected traits using unsupervised learning (ref 40)
- traditional methods in image processing not generalizable, in pathogen detection and require strict conditions and custom workflows to work [2]
- Current research Focus on image sensing, rather on computation which is new bottleneck. Imaging technology has advanced, but there is lack of robust analysis and computation tools. [bauckhageDataMiningPattern2013]
- Sensors generate huge amount of data, therefore algorithms need to be designed to deal with Terabytes of data. Furthermore, low cost computing units have limited bandwidth, battery and sensor capabilities indicating that algorithms need to take this in to account. Low complexity and efficient algorithms that can run on resource constrained tech is needed [bauckhageDataMiningPattern2013]
- Mobile applications can be used and run processing without need for external network, cyberinfrastructure. However there are not many applications available for this [muller-linowPlantScreenMobile2019]. Image analysis tools are often too expensive to run on these devices due to the constrained resources, however they are much cheaper and therefore can be exploited. In particular low complexity efficient systems, which generalize to various genotypes, and phenotypic measurements are needed. In particular the development of efficient low resource intensive machine modelling and deep learning techniques can be made.
- preprocessing technique choice is also needed, for example image segmentation methods can be costly, and a suitable method depends on various factors. Certain methods have cheaper cost, but less accuracy [muller-linowPlantScreenMobile2019]. In developing low-cost image analysis and preprocessing tools for phenotyping it is necessary to weight the cost savings versus accuracy and precision. But bespoke image processing techniques don't lend well to generalizability and extendability.
- cost-effective solutions require carefully choosing sensors and data processing techniques to be most cost-effective [30]
- automated systems not built for general solutions but for specific species of plants [6]

### **Current applications**

- Often combined with machine learning algorithms
- Cheap Raspberry PI with RGB camera [26]
- Autonomous robots with sensors and cameras [22]
- use off the shelf commercial software like microsoft xbox360 kinect [14]

### **What affects a low-cost automated greenhouse cost**

Points from this paper [11]

- Image preprocessing pipeline
- open source software
- hardware
- amount of labour and time needed for analysis
- energy cost of technology
- amount of data generated, where is it stored, how frequently? [5]

### **Challenges of low-cost automated greenhouses**

- Many phenotyping systems existing are commercially developed under patent so hardware and software can not be modified. [13]
- Current applications use moving conveyor belt (benchtop) [13]
- or conveyer belt [13]
- or moving robotic arm to move sensors around to get high-throughput measurements.
- Robot solutions not cheap for farmers [22]
- large amounts of data need to be collected, and processed. Existing phenotyping platforms exist, but they are expensive and have restricted running environments.
- Many systems developed for laboratory systems, but not extended for real applications. Standardised off the shelf equipment, open-source software allow modular and extensible plus affordable solutions [17]
- Custom data collection schemes need to be made very little standards
- lack of robust methods to analyse plant images
- low cost moving track systems limited in phenotypical, genotype and management types are supported [29] suggesting need for low-cost systems that can support a wider variety of genotypes and phenotypical traits.

Fill out the conclusion section.

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